GPU Sharing Mechanisms for Multi-Tenant Inference Serving

Group A: Search Execution and Evidence Report

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Abstract

This intermediate deliverable records execution of the submitted SLR protocol and prepares traceable evidence for the final CEP report. The 11 October IEEE/Springer checkpoint contains 689 formal query occurrences and 21 accepted studies. A 14 October extension supplied 23 ACM/ScienceDirect export files with 3,727 parsed occurrences. Those new records have not been fully screened or integrated into a revised review flow; the final included set and four-database PRISMA counts are pending. Run-count discrepancies, retrieval and extraction gaps remain explicit. The paper set and conditional synthesis are provisional; the completed four-database SLR and final CEP submission are not claimed.

Status: evidence checkpoint, not final SLR. The publication window is 1 January 2020–11 October 2026 inclusive. One AI reviewer conducted screening and appraisal at the user's direction; the listed students are the project authors, not claimed independent reviewers. Red text identifies statements, counts or conclusions that must be revisited when the remaining search reconciliation, screening, QA and extraction work is completed.

Contents

1 Purpose and requirements **4**

2 Protocol adoption and deviations **4**

3 Review design and operational definitions **4**
3.1 Eligibility decisions and source type 5

4 Database-specific search implementation **5**

5 Selection, identity and PRISMA accounting rules **7**

6 Quality appraisal and extraction instrument **8**

7 Search execution and provenance **9**

8 14 October ACM and ScienceDirect extension **9**

9 Screening checkpoint and reconciliation **11**

10 Current accepted primary studies **13**
10.1 Source-anchored finding register 14

11 Review of six supplied full texts **16**

12 Quality appraisal and extraction controls **17**
12.1 Section 14: data extraction status after the extension 17

13 Provisional answers to the research questions **18**
13.1 RQ1: mechanism and control boundaries 18
13.2 RQ2: outcome definitions and comparability 18
13.3 RQ3: operating conditions 18
13.4 RQ4: trade-offs and counterevidence 18
13.5 RQ5: conditional engineering choices 19

14 Sensitivity, limitations and continuation **19**

15 Evidence package and institutional requests **20**

A Study-level evidence cards **23**
A.1 GA-S001: Real-time, Work-conserving GPU Scheduling for Concurrent DNN Inference 23
A.2 GA-S002: Serving Heterogeneous Machine Learning Models on Multi-GPU Servers with Spatio-Temporal Sharing 24
A.3 GA-S003: PipeSwitch: Fast Pipelined Context Switching for Deep Learning Applications 25
A.4 GA-S004: Serving DNNs like Clockwork: Performance Predictability from the Bottom Up 26
A.5 GA-S005: Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference 26

A.6 GA-S006: StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow . . . 27
A.7 GA-S007: KRYPTON: Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception . . . 28
A.8 GA-S008: SIRIUS: Colocating ML Inference and Training with Fast GPU Memory Handover . . . 29
A.9 GA-S009: Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads . . . 29
A.10 GA-S010: ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments . . . 30
A.11 GA-S011: Orion: Interference-aware Fine-grained GPU Sharing for ML Applications . . . 31
A.12 GA-S012: GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform . . . 32
A.13 GA-S013: KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers . . . 32
A.14 GA-S014: XSched: Preemptive Scheduling for Diverse XPUs . . . 33
A.15 GA-S015: Paella: Low-latency Model Serving with Software-defined GPU Scheduling . . . 34
A.16 GA-S016: Optimizing High-Throughput Inference on Graph Neural Networks at Shared Computing Facilities with the NVIDIA Triton Inference Server . . . 35
A.17 GA-S017: VastPipe: A High-Throughput Inference System via Adaptive Space-Division Multiplexing for Diverse Accelerators . . . 35
A.18 GA-S018: ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs . . . 37
A.19 GA-S019: Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs . . . 38
A.20 GA-S020: iGniter: Interference-Aware GPU Resource Provisioning for Predictable DNN Inference in the Cloud . . . 39
A.21 GA-S021: InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems . . . 40

1 Purpose and requirements

The engineering problem is to share a physical GPU among inference tenants while balancing useful work and density against predictable latency and isolation. The assigned alternatives are time-slicing, MPS, MIG and API remoting/virtualization, including hybrids. The unit of technical comparison is local sharing on one GPU. A study containing cluster experiments qualifies only when local inference-sharing evidence is identifiable; cluster-wide GPU counts are never substituted for local occupancy.

The submitted `report/SLR_protocol.tex` and `report/SLR_protocol.pdf` are the protocol baseline identified by the user. The protocol contents are preserved. Query deliverables reside in the root `report/` folder; Group A evidence remains under `groups/group-a/evidence/`. The final assignment still requires 15–20 unique peer-reviewed primary studies, at least 15, a 12-point IEEE-cited LaTeX report, and approximately 20–30 pages of main content. Meeting a provisional paper count does not complete searching or the final deliverable. No additional intermediate-page requirement was supplied.

The five questions ask which mechanisms are evaluated (RQ1), how outcomes are measured and compared (RQ2), which operating conditions change results (RQ3), what trade-offs and gaps remain (RQ4), and what conditional choice is supported (RQ5). All five map to extraction and the claim register, rather than to an unconditional ranking of mechanisms.

2 Protocol adoption and deviations

The Group A execution copy adopts the submitted protocol. Amendment GA-A001 extends the cutoff from 27 September to 11 October 2026. GA-A002 records the user's instruction that Codex conduct screening and quality appraisal as a single AI reviewer. There is no independent second reviewer, inter-rater agreement statistic or adjudicated human consensus. GA-A003 permits supplementary full-text work while unavailable database coverage remains open. GA-A004 records the actual public-interface execution and metadata capture in place of signed-in publisher exports. None changes the inclusion criteria to meet a target count.

Eligibility requires an English peer-reviewed original journal or full conference report, a local sharing/partition/process/API mechanism, explicit concurrent inference or justified mixed inference evidence, and sufficient configuration/outcome detail. Training-only, cluster-only, surveys and vendor claims are excluded from the primary set. Unretrieved reports are pending access outcomes, not assessed full-text rejections. Related conference, journal and preprint reports are one underlying study until an independent contribution is established.

3 Review design and operational definitions

This report is the execution record for the adopted Group A protocol, not a second protocol. The decision unit is a unique evaluation of concurrent inference sharing on a physical GPU. An imported search hit is an *occurrence*; matching occurrences are linked to a canonical bibliographic record; a publication version is a *report*; and related reports can describe one *study*. These units are kept separate in every table and flow count. In particular, a server-level GPU saving or an eight-GPU placement outcome cannot be presented as a one-GPU utilization measurement merely because the same paper also evaluates local sharing.

RQ	Analytic question	Evidence needed
RQ1	Which temporal, MPS, MIG, API/runtime and hybrid controls were evaluated?	Mechanism, control layer, local tenant arrangement and comparator.
RQ2	Which outcomes and experimental methods make results comparable?	Metric definition, units, percentile/SLO, denominator, measurement window and baseline.
RQ3	Which workloads, hardware and configurations change the outcome?	GPU/SKU, profile/quota, model and batch, arrivals, co-tenant mix and resource pressure.
RQ4	What engineering costs, isolation properties and contradictory observations are supported?	Within-paper trade-offs, limitations, resource/fault/security experiments and counterexamples.
RQ5	Under which operating requirements is a choice supported?	Conditional synthesis with direct/indirect evidence and explicit unresolved conditions.

The main window is 1 January 2020 through 11 October 2026 inclusive. The publisher's broad 2020–2026 year filter is a retrieval aid; exact dates near the cutoff require checking at the report level. An older seminal exception requires a documented reason in the eligibility record. A result published after the cutoff is not made eligible by appearing in a later export. The exact-date decisions for potentially relevant 2026 reports in the new exports remain to be completed.

3.1 Eligibility decisions and source type

An included primary study must be an English peer-reviewed original journal or full conference report with identifiable concurrent inference evidence, a relevant local GPU-sharing control, and enough configuration and outcome detail to evaluate the research questions. Mixed inference/training studies can contribute only when the inference outcome is separable. Surveys and vendor descriptions may explain terminology but cannot supply an included primary experiment. Reports that cannot be retrieved remain access cases.

At full text, the adopted protocol records one principal reason, with secondary notes where helpful. E1 denotes no inference-serving evidence; E2 no relevant single-node multi-tenant sharing; E3 a mechanism outside scope; E4 training-only; E5 cluster orchestration without separable local sharing; E6 no peer-reviewed original primary report; E7 insufficient technical or outcome detail; E8 a duplicate or less complete report of the same study; E9 a date/language violation without exception; and E10 another written reason. A code is assigned after reading the relevant full report, not inferred from the publisher hit count. The final distribution of full-text reasons will be recalculated after the new reports are assessed.

4 Database-specific search implementation

The four NED-listed platforms cover engineering journals, computing proceedings and related systems literature, but their interfaces differ in fields, syntax and export behavior. The submitted protocol's common concept block combines GPU terminology with inference/model-serving

terminology; the mechanism block covers the assigned families. The recorded executed strings, rather than a retrospectively improved expression, define what was searched. The old five runs and the five later successful runs are separate dated executions. Blocked pilots are retained as attempts with no countable retrieved population.

Platform and family	Executed expression structure	Field/filter and consequence
IEEE MPS, MIG, temporal, API	`(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND` one family block.	All metadata; 2020–2026. The public implementation was not the planned field-qualified Command Search.
Springer master	Same GPU/inference blocks, with all four mechanism families in one OR block.	General search; English, Research or Conference Paper, 2020–2026. Full-text/reference indexing broadens irrelevant matches.
ACM master	Same common block and combined mechanism block, entered in Advanced Search.	All Fields, E-Publication Date 2020–2026; logged Full-Text Collection. Export page overlap/coverage is unresolved.
ScienceDirect temporal, MPS, MIG, API	`(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND` one family block per run.	The master exceeded the connector limit. The fallback omits DNN/LLM and is not equivalent to the other platforms' common block.

The exact entered and displayed queries, source URLs, observed hits and export receipts are retained in `search_runs.csv`, `search_runs_received.csv` and `export_intake.csv`; this table describes the logic, not a substitute for those verbatim strings. The ACM two-file overlap and the ScienceDirect MPS re-export show why a parsed-row count is not itself evidence of complete page coverage. The temporal URL's Open Access facet is documented separately from the unfiltered count observed later. The final search-coverage statement depends on the ACM export decision, verified filter provenance and any justified amendment; no missing-hit number is imputed.

A 17 October provenance review of an uploaded zip confirmed that all 23 included export files match the receipt ledger by hash and size, and found no screenshot or page-range record in that zip. The zip's file times appear to reflect a later copy and cannot establish the original export times. The helper's separate folder and the live PC directory have not been checked in this review. The two ACM files contain 829 and 1,000 entries with 442 repeated exact keys; overlap suggests page ranges may have intersected, but neither the original ranges nor any number of omitted results is known. The team must record a dated decision to re-export ACM in traceable chunks or retain this coverage limitation. The original run log is preserved, with its received copy differing only by a final CRLF. No query, log or export has been retrospectively changed to reconcile the site counts.

5 Selection, identity and PRISMA accounting rules

Every imported occurrence retains its source file, run and stable ID. An exact DOI or key can propose a link, while a normalized title with a different DOI triggers version review rather than automatic deletion. A conference paper, journal extension and preprint can be several reports of one study. The most complete peer-reviewed report is preferred for claims, while the related versions remain in the history. Whole-proceedings volumes and books in the ACM export need an explicit non-primary-record treatment; their bibliographic keys do not make them individually evaluated papers.

The 11 October checkpoint demonstrates the arithmetic for its captured subset: 689 formal occurrences minus 10 duplicate occurrences plus 20 selected supplementary records yields 699 records considered; 575 earlier-stage exclusions and 96 pending/not-retrieved cases leave 28 assessed reports; seven report-level exclusions leave 21 unique accepted studies. This is a dated checkpoint, not the final four-database flow. In the later import, the 515 exact DOI/key links are first-pass occurrence classes; title/version and cross-source relationships remain to be adjudicated. Neither the 3,727 new occurrences nor the 2,615 assisted/individual first-pass exclusions can be placed directly in the final PRISMA boxes without settled decisions.

Stage	Reconciliation rule and required ledger evidence
Identification	Sum imported occurrences by database/run and supplementary route; retain source file and original display observations separately.
Before screening	Count adjudicated duplicate occurrences and any other removals from named occurrence-to-canonical links; do not remove a report merely for sharing a study ID.
Title/abstract	Count canonical records screened and exclusions with reasons; keep title-only and abstract-informed decisions distinct.
Retrieval	Map records advanced to reports sought; record each not-retrieved report and access attempts without treating it as an assessed exclusion.
Full text	Count assessed reports and one principal exclusion reason per excluded report; link multiple included reports to unique studies.
Inclusion	Show both included reports and unique studies, with study-to-report identity reconciled.

The final four-database PRISMA flow, its reason totals and included-study count must replace the interim figure after the new occurrence ledger is adjudicated. The flow cannot be completed by choosing a convenient difference between a displayed publisher total and an export count. A publisher interface observation and the set of files actually received remain separate provenance facts.

6 Quality appraisal and extraction instrument

Quality scoring is applied to eligible evidence after full-text assessment; it is not a threshold that creates eligibility. Each criterion receives 0, 0.5 or 1, with an individual rationale and source anchor. High (8–10), moderate (6–7.5) and low (below 6) are reporting/interpretability bands. A paper's score is not a security certification, replication claim or proof that its baseline is suitable for all uses.

Item	Question scored from the report
Q1	Is the engineering question and tenant setting explicit?
Q2	Is the mechanism and its configuration sufficiently described?
Q3	Are hardware, software and resource environments identifiable?
Q4	Are workloads, arrival patterns and co-tenant conditions described?
Q5	Is the comparator appropriate and its resource budget clear?
Q6	Are metrics, units, percentiles and denominators defined?
Q7	Are repetitions, warm-up, measurement windows and other controls adequate?
Q8	Does the cited figure/table/prose support the stated result?
Q9	Are validity limits and adverse cases discussed?
Q10	Can the setup be reproduced or at least critically interpreted?

The extraction form links a preferred report and study ID to source type/version, GPU generation and partition or quota, driver/runtime, model and tenant mix, batching/arrival process, comparator, metric and statistic, a bounded outcome, caveat and exact source location. It distinguishes a missing transcription (`NOT EXTRACTED`) from a checked absence in the report (`not reported`). A graph with unreadable pixels is not reverse-engineered into an exact value. The first 21 rows and the two preliminary new assessments are at different maturity levels; newly eligible studies require formal QA, version linkage and a complete anchored extraction before joining the synthesis.

Isolation field	What the observation can and cannot support
Performance interference	Measured latency/throughput under specified co-tenants; cannot alone prove a security boundary.
Resource and memory separation	Profile/quota or memory behavior and measured contention; a partition label alone does not establish an observed guarantee.
Adversarial security	Explicit hostile-tenant or access-control experiment, if reported; otherwise mark not established.
Fault containment	Explicit failure or fault-propagation experiment, if reported; ordinary workload fairness is not a substitute.

The source-linked cards in Appendix A give an auditable QA and extraction view for the current 21-study checkpoint. They are a compact reproduction of the Group A registers, not additional independently reviewed studies. Cards, scores and claims must be regenerated if eligibility, preferred versions or source-grounded extraction changes.

7 Search execution and provenance

NED's E-Resources page lists the four planned publisher platforms. The listing was checked on 11 October 2026: `https://eaklibrary.neduet.edu.pk/eakl/E-Resources.html`. It confirms platform availability in the list, not entitlement to each requested article. No university credentials or licensed full-text PDFs enter this public repository.

Source/run	Actual implementation	Observed hits
IEEE MPS	All metadata; years 2020–2026	25
IEEE MIG	All metadata; years 2020–2026	23
IEEE temporal	All metadata; years 2020–2026	45
IEEE API	All metadata; years 2020–2026	9
Springer master	General search; English; research articles/conference papers;2020–2026	587
ACM pilot (11 October)	Search blocked by challenge loop	Unknown; no import at that checkpoint
ScienceDirect pilot (11 October)	Search page unavailable	Unknown; no import at that checkpoint

The common block is GPU or graphics-processing-unit AND inference, model serving, deep learning, DNN or LLM. IEEE uses one family block per run. Springer uses all family blocks in one OR group. Exact strings, filters, source URLs, dates and export limitations are in `search_runs.csv`. The IEEE field implementation differs from the planned field-qualified Command Search and is disclosed, not described as an equivalent validated command. Research type/language and precise 2026 cutoff checks are manual where facets are unavailable. No accepted study is dated after 2025.

All 30 Springer result pages (20 hits/page,7 on the final page) and all IEEE pages were captured, yielding 587+25+23+45+9=689 occurrences. These are rendered-result metadata imports, not publisher-generated RIS/CSV exports. Springer export redirects to sign-in; IEEE results export requires personal sign-in and public citation export timed out. The imports retain IDs, database/run, page, title, DOI/URL and displayed date/item metadata. Clock time was not retained and is marked unrecorded.

Springer's broad index includes reference/full-text matches and unrelated acronym expansions such as welding MIG. Search hits therefore do not imply eligible GPU studies. Seed checks recovered ParvaGPU and KRISP in IEEE; papers in other publisher proceedings required supplementary discovery at this checkpoint. Supplementary records are selected named-paper lookups checked after adoption, including submitted seeds and related systems. They are not an exhaustive search-engine result export or completed backward/forward citation search. Their follow-up query strings are reproducible suggestions, not invented historical exact queries.

8 14 October ACM and ScienceDirect extension

The operator supplied 23 database export files and a search log from the named PC export directory. All 23 export SHA-256 hashes were checked against `export_intake.csv` on 16

October and matched. A separate 17 October check of an uploaded repository/export-folder zip likewise found 23/23 matching hashes and byte sizes; this does not verify the live PC directory or a private backup, which remains unconfirmed. The 24th export-folder item is the log, not a database export. The original 6,302-byte log and 6,304-byte `search_log_received.txt` have identical text lines; the received copy adds one final CRLF. The source-file receipt and per-file parser counts are in `export_intake.csv`; the exact received log and structured run transcription are `search_log_received.txt` and `search_runs_received.csv`. The log records ACM's Full-Text Collection and all-fields query, four ScienceDirect family searches, the recorded 2020–2026 year filters, URLs and search times; export times are not recorded. The ACM log states that the run used NED library access, without recording the access mechanism; the operator reported NED VPN use for the ScienceDirect runs. Neither statement establishes entitlement to individual full texts. The protocol's exact 11 October cutoff still applies to individual reports.

Run	Displayed hits	Log export claim	Parsed entries
ACM master	1,600	1,603	1,829
ScienceDirect temporal	1,053	1,053	1,053
ScienceDirect MPS	517	563	563
ScienceDirect MIG	241	241	241
ScienceDirect API	41	41	41
Total of run values	3,452	3,501	3,727

The parsed total consists of 1,829 BibTeX and 1,898 RIS entry *occurrences*, before adjudicated deduplication. The ACM files contain 442 exact keys in both files and 1,387 distinct keys altogether. Of those distinct keys, 472 label whole-proceedings volumes rather than individual papers; their treatment must be recorded in screening. The site showed 1,600 in the saved run register and 1,603 according to the operator; the export contains 1,829 rows. The page-range overlap and completeness of the export are unverified. The numbers represent site observations, parsed rows and keys, respectively, and are not interchangeable. ACM exports used the site's Export button in one session with different page ranges, according to the operator.

ScienceDirect MPS contains 563 parsed entries, including 63 from the separately re-exported `SSD_MPS_06.ris`; 561 distinct DOIs occur in those files. The source of the earlier displayed count of 517 is unrecorded. Reopening the result page on 16 October showed 563 hits, a later observation that supports but does not establish the 14 October page state. The ScienceDirect temporal files contain 1,053 rows and 1,051 distinct DOIs; their `_010/_011` filename order was reversed by a naming error, without changing their retained contents. Its logged URL carries an `openaccess` facet absent from the written filter description. On 16 October, the operator observed 298 hits with that facet and 1,053 without it. The 1,053 exported entries are consistent with the unfiltered set, but the filter state at export time cannot be established from this later check. No original log or export is altered to force these counts to agree.

ScienceDirect rejected the intended master string under its reported eight-connector limit, so the four family queries omitted the `DNN OR LLM` terms in the inference block. The received log preserves the exact queries. Family splitting follows protocol Section 5, but dropping terms from the shared inference block is a deviation whose effect on recall is unmeasured. The team has not decided whether to rerun or formally enter an amendment in the protocol register. Any additional content-type or language filter is not recorded. The 2026 year facet alone does not enforce the 11 October cutoff. The five 14 October completed runs are recorded separately from the 11

October blocked pilots. Each log names the first export file for its run; `export_intake.csv` maps all 23 files.

The reproducible import assigns `GA-N00001`–`GA-N03727` in `new_export_occurrences.csv`. Automated DOI/title matching flags nine occurrences against the old 689 formal records, 537 against earlier new occurrences, and twelve candidate links to already accepted study metadata; these classes can overlap and require report-version adjudication. The subsequent reproducible first pass in `new_screening_triage.csv` links 427 exact DOI repeats and 88 exact ACM BibTeX-key repeats, records 2,615 title/abstract exclusions (six individually documented and 2,609 rule-assisted), retains ten priority retrieval leads, separately records two assessed eligible full reports, and leaves 585 topical or ambiguous occurrences for individual title/abstract review. These disjoint occurrence classes sum to 3,727. Rule-assisted exclusions need sensitivity review, 585 occurrence decisions need individual review, and duplicate/version links need adjudication before the final flow is calculated. The two assessed reports have source-anchored preliminary extraction in `new_fulltext_assessments.csv`, but still need formal QA and version linking before integration into the 21-study synthesis and final flow. A retrieval lead is not an included study. The 21 accepted studies and their source-anchored extraction below remain the prior checkpoint.

9 Screening checkpoint and reconciliation

Every formal occurrence receives a record ID. Exact DOI/URL and normalized-title checks link duplicate occurrences without deleting provenance. Bibliographic report IDs identify candidate reports; study IDs identify accepted underlying studies. The REEF journal extension is selected for GA-S001; its less complete conference report is a linked E8 companion and adds no study.

Checkpoint quantity	Count
Formal query occurrences	689
Duplicate occurrences	10
Unique formal records	679
Selected supplementary records	20
Excluded before full text	575
Full texts assessed	28
Full-text exclusions	7
Accepted unique studies	21
Pending / not retrieved	96

The reconciled captured subset is: 689-10+20=699 records; 699 equals pre-full-text exclusions plus assessed reports plus pending records. Assessed reports equal included reports plus full-text exclusions. The later ACM/ScienceDirect occurrence population is outside these 11 October arithmetic identities. This 11 October checkpoint is intentionally not a completed PRISMA 2020 flow and must be replaced by a reconciled four-database flow. Reports sought include pending retrieval attempts, but complete formal-database retrieval and final study-version resolution remain open.

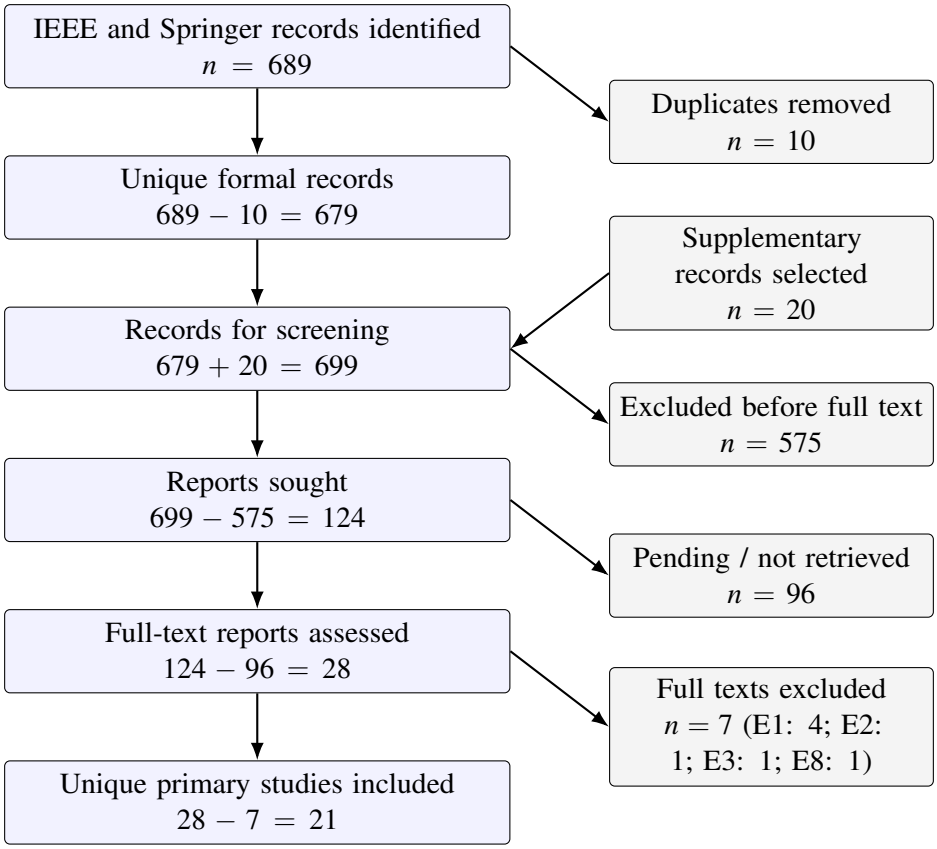


Figure 1: Interim PRISMA 2020 stage mapping for the reconciled 11 October captured subset. The later 3,727 ACM/ScienceDirect occurrences are outside this flow. E1–E8 reason definitions and report-level decisions are in `rejected.csv`. This is not the completed four-database flow.

Title decisions are first-pass relevance judgments, explicitly labelled as title stage. Abstract decisions cite publisher scope; full-text decisions identify a section or experiment. Training and inference are not interchangeable: PipeSwitch and SIRIUS are retained for explicit inference outcomes, whereas TGS is rejected for training-only evidence. The Fermilab paper was retained after its multi-model slice experiment was inspected, rather than merely because it mentioned MIG. DIPPM was rejected because profile prediction alone does not test concurrent tenant sharing.

Full-text rejection examples include DIPPM (no concurrent tenant comparison), the accelerator-pool vRAN simulation and the low-latency FEC offloading paper (no inference evidence), DRMaestro's Rodinia workload evaluation (no serving inference), CalcGraph (assigned GPU partitioning left to future work), and the mediated-device passthrough benchmark (HPC applications without inference). The complete reasons and stages are in `rejected.csv`. Pending files have no fabricated technical results or QA scores.

10 Current accepted primary studies

The following studies are accepted for this intermediate evidence set. The study list, preferred reports and QA totals may change after the new search population is screened and version links are settled. KRISP is a scope-boundary inclusion under protocol 7.2 because it evaluates local spatial resource control and inference workers; its emulated AMD feature and non-NVIDIA baseline are explicitly isolated in synthesis. No vendor document or preprint-only unpublished work counts toward this table.

Study	Report/year	Mechanism / scope	QA
GA-S001	REEF 2025 [1]	Reset-based temporal preemption + dynamic kernel padding; NVIDIA binary instrumentation	8.5
GA-S002	GPULET 2022 [2]	MPS + temporal	8.5
GA-S003	PIPESWITCH 2020 [3]	Temporal context switching	9.0
GA-S004	CLOCKWORK 2020 [4]	Temporal exclusive scheduling	8.5
GA-S005	TORPOR 2025 [5]	API remoting + temporal swapping	8.5
GA-S006	STREAMBOX 2024 [6]	API/runtime sandbox virtualization	8.0
GA-S007	KRYPTON 2025 [7]	Kernel API virtualization + hybrid	9.0
GA-S008	SIRIUS 2025 [8]	Temporal + MPS memory handover	8.5
GA-S009	TALLY 2025 [9]	CUDA API interception + priority	9.0
GA-S010	PARVA 2024 [10]	MIG + MPS	8.0
GA-S011	ORION 2024 [11]	CUDA API interception + temporal	9.0
GA-S012	GSLICE 2020 [12]	MPS adaptive provisioning	9.0
GA-S013	KRISP 2023 [13]	Spatial partition/API control (AMD)	8.0
GA-S014	XSCHED 2025 [14]	Temporal API interposition	9.0
GA-S015	PAELLA 2023 [15]	Temporal software scheduling + MPS comparison	8.5
GA-S016	GNNTRITON 2024 [16]	MIG + concurrent model serving	8.5

Study	Report/year	Mechanism / scope	QA
GA-S017	VASTPIPE 2025 [17]	MPS adaptive quotas; separate AMD CU-mask stratum	6.5
GA-S018	ARKGPU 2023 [18]	CUDA API interception + NVML monitoring and admission throttling	7.0
GA-S019	DYNAP 2025 [19]	MIG dynamic placement/reconfiguration with reconfiguration-last policy	7.0
GA-S020	IGNITER 2023 [20]	MPS active-thread quotas + interference-aware batch/provisioning and shadow process	8.5
GA-S021	INFERFAIR 2024 [21]	Temporal SLO-aware scheduling + feedback goodput balancing; MPS comparator	7.5

USENIX proceedings PDFs are official published copies. ACM/IEEE author or camera-ready copies are linked to verified proceedings metadata and do not become separate preprint studies. ParvaGPU's author manuscript is provisional for publisher-version comparison; its reported runtime-version anomaly remains flagged. The supplied REEF journal conversion establishes a more complete implementation/evaluation; it is now the preferred report for GA-S001. The conference report is retained as a linked companion. Accessible copies of Orion, GSLICE, KRISP, Tally and Paella avoid unnecessary institutional download requests.

10.1 Source-anchored finding register

Study	Bounded finding	Source location
GA-S001 [1]	GV100 PyTorch DISB-C: Orion p 99 normalized latency 1.17x vs Reef below 1.05x; normalized total throughput 0.97x vs 1.14x(Fig 29). TensorRT comparison trades throughput: mean 1.21x Triton vs 1.11x Reef(Fig 28). All ratios retain model-solo normalization; do not mix engines/AMD strata.	Article pp 1:23-1:24,1:31-1:34; sections 6.2,6.6,7; Figs 28-29; Markdown lines 1111-1180,1600-1820
GA-S002 [2]	Local gpulets support interference-aware provisioning; cluster throughput is a different outcome.	PDF p 10 Table 2/3; section 5; local profiling section 3
GA-S003 [3]	Fig 7: inference preemption latency reduced by up to 38.2ms.	PDF pp 11-13; sections 6-6.2; Fig 7
GA-S004 [4]	Fig 5 measures goodput satisfying SLO, rather than raw request throughput.	PDF p 10; section 6.1; Fig 5
GA-S005 [5]	BERT Table 4: native 42ms, remote 43ms, PCIe swap 144ms, NVLink swap 45ms.	PDF pp 10-11; Table 4; sections 7.2-7.3
GA-S006 [6]	Table 2 evaluates runtime sandbox workflows on shared GPUs.	PDF p 10 onward; section 9; Table 2

Study	Bounded finding	Source location
GA-S007 [7]	Inference-specific sharing comparisons; section 2.2 tests fault propagation.	PDF p 10 Tables 1-2; section 5; section 2.2 Fig 3
GA-S008 [8]	Inference p 99/SLO outcomes accompany training progress; separate TTFT/TBT LLM case.	Section 6.1 Fig 10; section 6.6
GA-S009 [9]	At 10% idle time: normalized throughput Tally 1.19 vs TGS1.65.	PDF pp 9-11; sections 5-5.3; Fig 6
GA-S010 [10]	Fig 7 separates internal/external fragmentation; Fig 5 counts GPUs.	PDF p 8 onward; sectionIV; TableIV; Figs 5/7
GA-S011 [11]	Interference-aware scheduling compares inference latency and BE throughput.	PDF pp 8-9; sections 5.3-6.1; Tables 1/3
GA-S012 [12]	GUE measures efficiency; shadow instance loading takes about 12s in Fig 15.	PDF pp 8-12; section 4; Table 3; Fig 15
GA-S013 [13]	Kernel-wise right-sizing evaluates throughput and p 95 inference latency.	PDF pp 9-10; sectionsV-VI; TableIII
GA-S014 [14]	Fig 15: relative p 99 1.53/1.51/1.07 respectively versus solo.	PDF pp 14-15; section 8 Case 3 Fig 15; section 9
GA-S015 [15]	p 99 latency and throughput under bursty model mixes; Fig 12 includes MPS.	PDF pp 9-12; section 7 Table 3/Figs 11-12; section 8
GA-S016 [16]	Fig 10: thrashing after two models/20GB or five/40GB.	HTML Multi-model Scaling/Fig 10; Triton implementation; limitations
GA-S017 [17]	A100 VGG16+GoogLeNet: throughput 2.7x no MPS and 1.5x default MPS; not a universal average or p 99 result. NLP/CV mean 2.43x no MPS, 1.45x default MPS. MI100 mean 1.89x no mask, 1.24x default mask.	Published pp 457-459; sections 6.7-6.9; Figs 14-16; uploaded Markdown lines 1150-1290
GA-S018 [18]	Section 5.2 reports all 30 tested pairs satisfy the 2x-solo p 99 QoS target with ArkGPU. Low-contention pairs can lose about 5% QPS from throttling. Four-GPU app-mix goodput gains 94.98% vs k 8s-native and 38.65% vs MPS are separate system-level results.	Published pp 313-319; sections 4.3,5.1-5.4; Table 3; Figs 14-21; Markdown lines 920-1360
GA-S019 [19]	Single-A30 batch experiments: smaller partitions increase concurrency but can increase mean JCT; not online p 99 evidence. Fig 8 broader job set reports 39.03% lower makespan vs StaCon,12.14% vs MISO and 14.7% higher normalized STP vs StaCon. Prediction-error sensitivity:15% perturbation adds 8.09% makespan.	PDF pp 11-16; sections 4.1-4.3.2; Table 2; Figs 8-13; Markdown lines 700-865

Study	Bounded finding	Source location
GA-S020 [20]	Local four-model latency prediction error 1.53%-5.02%(Fig 13). Fig 14 deployment: iGniter 6 GPUs/0 SLO violations vs gpu-lets+8/3,FFD+5/10,GSLICE+6/3;25% lower hourly instance cost vs gpu-lets+, not 25% higher local utilization. Fig 17 shows an initial p 99 violation before shadow-process recovery.	Published pp 820-824; sections 4.2,5.1-5.4; Figs 13-18; Markdown lines 938-1245
GA-S021 [21]	Local Table 3 load 3:1:2: unfairness 1.05 vs Clockwork 1.50; SLO goodput 524.29 vs 457.24req/s. Other load ratios give smaller differences. Local unfairness is max/min model slowdown(Eq 2), not p 99 or security. Four-GPU MAF outcome kept separate.	Published pp 17-18; sections 5.1-5.4; Table 3; Eq 2; Markdown lines 718-899

11 Review of six supplied full texts

Five reports add unique studies: VastPipe, ArkGPU, Dyna-P, iGniter and InferFair. REEF's journal report explicitly extends OSDI2022 with a complete NVIDIA implementation, additional kernel handling and newer comparisons; the journal is preferred within GA-S001. Six previously pending reports are now assessed; the earlier conference report becomes a full-text E8 companion. There are 21 accepted studies,28 assessed reports including 7 full-text exclusions, and 96 pending access cases. Search occurrence and discovery counts are unchanged.

The source files are Markdown conversions. Two-column interleaving, missing plot pixels and damaged mathematical symbols prevent exact extraction of some equations, CDFs, error bars and configurations. Only readable labelled tables, captions and explicitly stated numeric results enter the register. Missing values are not interpolated. The receipt ledger records filenames and SHA-256 hashes; licensed source contents are not published.

Local GPU strata determine inclusion. VastPipe's NPU-cluster outcomes, ArkGPU's four-GPU goodput, Dyna-P's multi-GPU aggregates and cluster simulation, iGniter's fleet cost, and InferFair's four-GPU trace are distinguished from local results. ArkGPU qualifies through the explicit mixed inference/training exception; transfer is limited to priority inference protection. Dyna-P's job completion and makespan do not establish online p 99. InferFair's unfairness is a relative slowdown ratio and does not establish adversarial isolation.

Journal REEF's same-engine GV100 PyTorch comparison reports DISB-C p 99 below 1.05 times solo versus 1.17 for Orion, alongside separately normalized total throughput. Its TensorRT comparison also reports lower mean normalized throughput than Triton; preserving this counterexample prevents a universal superiority claim. iGniter's initial p 99 violation followed by shadow-process correction likewise qualifies any steady-state SLO statement. Source locations and QA rationales are recorded per report.

At the stricter QA threshold 8, VastPipe, ArkGPU, Dyna-P and InferFair are omitted; their mechanism-specific conclusions weaken. At the protocol low-quality threshold 6, none is omitted. This is a reporting-quality sensitivity, not a pooled causal result. The set now exceeds the

protocol's 15–20-study target range; retain all eligible studies at this checkpoint and document any final narrowing transparently rather than removing evidence to fit a count.

12 Quality appraisal and extraction controls

Q1–Q10 cover tenant question, mechanism/configuration, environment, workload/arrivals, comparator fairness, metric clarity, measurement controls, result support, validity limits and reproducibility/interpretability. Scores are 1 adequate,0.5 partial and 0 absent; totals 8–10 are high,6–7.5 moderate and below 6 low. The single-reviewer scores, concise rationales and source anchors are in `quality.csv`. Scores assess reporting and interpretability; they do not prove unbiased results, system security, reproducible artifacts or complete searching.

Current totals range from 6.5 to 9.0. Four newly included studies score below 8; their configuration or measurement controls and missing plot data require caution. Environment and measurement-control fields are frequently partial. No authors' experiments or code artifacts were rerun. Artifact availability alone is not experimental replication. The extraction register captures source/version, mechanism, hardware/runtime, tenant workload, comparator, a bounded finding, caveat and precise source location. Fields not yet fully transcribed are labelled NOT EXTRACTED, distinct from a source's not-reported value. A further detailed extraction pass is needed before final numerical comparisons.

Isolation observations remain four different columns: performance interference, resource/memory separation, adversarial security and fault containment. Most current findings establish workload performance rather than comparative security or fault containment. The KRYPTON fault experiment is specifically identified; it does not validate all mechanisms or all GPU generations.

12.1 Section 14: data extraction status after the extension

The extraction and appraisal matrices for the 21 accepted studies were prepared from the previously assessed source texts. No new ACM or ScienceDirect occurrence has been added to these matrices solely on the strength of bibliographic metadata. Two ACM full proceedings reports have now been assessed directly from author-hosted copies: Pantheon (MobiSys 2024) evaluates concurrent inference scheduling on Jetson GPUs and reports deadline misses of 0.39–1.10% in its three application mixes (PDF section 5.1, Fig. 9); Jormungandr (SYSTOR 2024) profiles one-A100 multi-model MIG/MPS/batch configurations but evaluates allocation on an eight-A100 cluster (PDF sections 3.2 and 8, Figs. 3 and 6–7). These are provisionally eligible, with source-anchored fields and transfer limits in `new_fulltext_assessments.csv`; formal QA, report-version linkage and synthesis integration remain open. The ten other priority leads and any of the 585 unresolved occurrences advanced after individual review need full texts, eligibility decisions, study/version links, quality appraisal and source-anchored extraction. Title/abstract exclusions have no full-text or extraction claims. The workbook's `Extension`, `NewQueue`, `Triage` and `AssessedNew` tabs show the run receipt, immediate review queue, occurrence-level first pass and these two full-text assessments; the source CSV retains all 3,727 records. Thus Section 14 is an auditable status through extraction, not an assertion that extraction is complete for all newly received records.

13 Provisional answers to the research questions

The following RQ answers are a synthesis of the current 21-study checkpoint. Their coverage and confidence must be checked against the final eligible set and completed extraction.

13.1 RQ1: mechanism and control boundaries

The set contains temporal scheduling, MPS resource provisioning, MIG plus MPS, and API/runtime interception or remoting. Hybrids are counted as studies once, not once per mechanism. Software scheduling may control task admission, kernel dispatch, preemption or memory handover; these are different operating boundaries. A process-sharing prototype, a kernel-intercepted container and a hardware slice cannot be assigned the same security meaning. The detailed mapping is in `extraction.csv` and `claims.md`.

13.2 RQ2: outcome definitions and comparability

Keep requests/second, SLO-compliant goodput, allocated GPU count, GPU busy time, SM occupancy, memory utilization and paper-specific efficiency ratios separate. Likewise p 95, p 98, p 99, average latency, time-to-first-token and time-between-tokens are not substitute measurements. A reduction in context-switch latency does not by itself establish p 99 end-to-end latency isolation. Normalize only within an explicitly comparable experiment and retain the original denominator and statistic.

Comparison check	Required condition before numerical comparison
Hardware	Same GPU/vendor/memory/profile or explicit indirect label; separate AMD emulation from NVIDIA mechanisms.
Workload	Same models/input sizes, batching, tenants and co-tenant mix; mixed training comparisons retain inference outcomes.
Load	Same open/closed-loop arrival process, rate, burstiness, warm-up and measurement window.
Comparator	Same resource budget; disclose reimplementations, wrappers and baseline adaptations.
Outcome	Same latency percentile/SLO and throughput denominator; occupancy and device-count reduction remain separate.

13.3 RQ3: operating conditions

Current evidence suggests that model memory demand, launch granularity, arrival burstiness, batching, priority mix and transfer path alter the sharing result. GPU generation constrains which resource boundaries are available. Local slice experiments are informative for memory pressure, while cluster scaling results require their own placement/network interpretation. Recommendation confidence falls when driver/runtime, quota/profile and experiment windows have not yet been completely transcribed.

13.4 RQ4: trade-offs and counterevidence

Stricter priority protection may trade lower-priority throughput for predictability. Remoting or swapping adds path-dependent transfer cost. Resource partitioning may reduce packing

flexibility. Fine-grained control can require compiler changes, API interception, driver hooks or baseline adaptations. These implementation costs must accompany reported gains. The source-anchored counterexample and parameter-specific results reside in the extraction register, avoiding an invented common percentage improvement across studies.

13.5 RQ5: conditional engineering choices

The following are provisional reviewer interpretations, not validated deployment instructions. The candidate/evidence links must be refreshed after final selection and QA sensitivity analysis.

Operating objective	Candidate to investigate	Evidence / unresolved condition
Priority-sensitive inference	Temporal/API scheduling with admission and memory controls	GA-S001, S003, S004, S008, S009, S011, S014; direct per-paper tests, indirect transfer.
Density with controlled workload mix	Tuned MPS provisioning	GA-S002, 012; check memory/compute contention and reconfiguration cost.
Slice boundary with concurrent models	MIG, optionally MPS inside a slice	GA-S010, 016; profile, packing and memory pressure; no universal isolation result.
Serverless remoting and elasticity	API remoting/runtime virtualization	GA-S005, 006,007; runtime compatibility and transport budget.
Strong adversarial/fault guarantee	No universal selection supported	Comparative evidence insufficient; retrieve security candidates and distinguish failure types.

14 Sensitivity, limitations and continuation

Removing QA<6 removes no current study, so that sensitivity test is uninformative. Removing provisional author-manuscript evidence weakens hybrid MIG and the recorded negative throughput comparison. Excluding AMD-only feature evidence limits cross-vendor conclusions. Removing all incompletely transcribed environment profiles would sharply reduce evidence; final recommendations require completing those fields. No pooled effect or universal winner is asserted.

The main limitations are unresolved ACM/ScienceDirect run-count and deduplication reconciliation, incomplete screening of newly supplied exports, single-AI-reviewer bias, inaccessible full texts, supplementary-selection bias, baseline and hardware heterogeneity, incomplete parameter extraction and unresolved report versions. The accessible set is concentrated in systems proceedings and older DNN workloads; LLM and adversarial isolation evidence requires further checking. A minimum paper count cannot offset these limitations.

Next, adjudicate export selection and cross-source duplicate links, screen the new records under the frozen protocol, continue public-access recovery, complete parameter extraction and rerun claims/sensitivity. The user supplied six of the ten requested papers; all six are assessed. The remaining four are unavailable and no further paper download is requested. Link any new report to its underlying study before adding a citation. Exact 2026 dates must be verified. Scope changes still require a documented amendment and instructor guidance; this report does not relax eligibility.

15 Evidence package and institutional requests

The evidence directory supplies search runs, imported records, duplicates, all decisions and reasons, accepted/rejected/pending views, retrieval lookups, QA, extraction, claims and checkpoint counts. `query_references.bib` contains only the 21 currently chosen primary reports, each cited here. The spreadsheet is a review view; CSV/JSON records remain the authority. Scripts validate linkage and reconciliation and regenerate derived views.

The unresolved access backlog is in `download_requests.csv`; there are zero active requests. `download_requests.md` records the six received conversions and four unavailable original requests. No additional paper download is requested. All requested publisher platforms appear on NED's list, but article-level access must be checked using university access. PDFs should be supplied privately through the conversation, not committed to the public repository. The supplied export files preserve a new occurrence ledger, while exact syntax, collection, facets and conflicting displayed/export totals remain recorded for reconciliation.

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A Study-level evidence cards

These cards reflect the 21-study checkpoint; regenerate them after the final study and report-version decisions. The item rationales below reproduce the Group A quality register. Bounded outcomes and caveats reproduce the extraction register. They are not a new independent appraisal or a substitute for the cited full texts.

A.1 GA-S001: Real-time, Work-conserving GPU Scheduling for Concurrent DNN Inference

Preferred report: ACM Transactions on Computer Systems, 2025; report GA-PX007. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	Reset-based temporal preemption + dynamic kernel padding; NVIDIA binary instrumentation
Environment	One MI50, 16GB/60CUs; i7-10700, 32GB RAM; Ubuntu20.04, ROCm4.3.0, TVM0.14.0. Separate one GV100, 32GB/80SMs; CUDA11.4, PyTorch1.12.0, TensorRT8.5.1. Driver version NOT EXTRACTED.
Workload	DISB A-E, uniform RT except Poisson E; closed-loop BE; ResNet152, VGG19, YOLOv3, BERT-base, GPT2 (128 tokens, GPT2 prefill). RT rates50%,100% or10% solo-throughput according to workload; Apollo/SVL trace. Separate LLM case batch1,100 decode tokens, MLC-LLM.
Comparator	RT-only; sequential priority; GPU streams. NVIDIA closed-source strata compare priority-enabled Triton/TensorRT and Orion/PyTorch without dynamic padding.
Bounded finding	GV100 PyTorch DISB-C: Orion p99 normalized latency1.17x vs Reef below1.05x; normalized total throughput0.97x vs1.14x(Fig29). TensorRT comparison trades throughput: mean1.21x Triton vs1.11x Reef(Fig28). All ratios retain model-solo normalization; do not mix engines/AMD strata.
Transfer limit	Explicit OSDI2022 extension; select journal as preferred report and retain GA-S001 once. Requires deterministic/stateless kernels; closed-source comparison omits padding. Memory assumed sufficient; bandwidth contention and input-shape profiling limits. Printed publication date differs from some issue-year indexing; recorded2025 follows supplied copy and is in cutoff either way.
Source anchor	Article pp1:23-1:24, 1:31-1:34; sections6.2, 6.6, 7; Figs28-29; Markdown lines1111-1180, 1600-1820

QA	Score	Source-based rationale
Q1	1	RT and BE sharing question explicit

QA	Score	Source-based rationale
Q2	1	Reset/preemption, padding and NVIDIA implementation described
Q3	0.5	GPU/CPU/RAM/OS/frameworks clear; driver missing
Q4	1	DISB mix, RT relative rates, arrivals and model/token configurations explicit
Q5	1	Same-engine closed-source comparisons and priority settings disclosed
Q6	1	Latency and throughput denominators explicit; p99 prose anchored toFig29
Q7	0.5	Solo reference averaged1000 runs; repeats/duration for all serving tests unresolved
Q8	0.5	Captions/numerical comparisons survive; full CDF/plot data absent
Q9	1	Section7 details statelessness, determinism, graphs, bandwidth and memory limits
Q10	1	Implementation and explicit engine constraints permit interpretation; experiments not rerun

A.2 GA-S002: Serving Heterogeneous Machine Learning Models on Multi-GPU Servers with Spatio-Temporal Sharing

Preferred report: USENIX ATC, 2022; report GA-PS002. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	MPS + temporal
Environment	RTX2080Ti 11GB; two/server; Ubuntu18.04; CUDA10.2; driver440.64; PyTorch1.10
Workload	Nine models; Poisson arrivals; per-model SLO
Comparator	Adapted SBP/Nexus; best-fit with/without interference
Bounded finding	Local gpulets support interference-aware provisioning; cluster throughput is a different outcome.
Transfer limit	Server-level gains cannot be treated as one-GPU utilization.
Source anchor	PDF p10 Table2/3; section5; local profiling section3

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	1	Environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls

QA	Score	Source-based rationale
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.3 GA-S003: PipeSwitch: Fast Pipelined Context Switching for Deep Learning Applications

Preferred report: USENIX OSDI, 2020; report GA-PS003. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal context switching
Environment	V10016GB or T416GB EC2; PyTorch1.3; CUDA10.1
Workload	Training batch32; inference8; RN152/InceptionV3/BERT; 100 runs
Comparator	Unoptimized/grouped/per-layer transfer
Bounded finding	Fig7: inference preemption latency reduced by up to 38.2ms.
Transfer limit	Average switch/inference latency is not p99; multi-VM throughput separate.
Source anchor	PDF pp11-13; sections6-6.2; Fig7

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	1	Measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.4 GA-S004: Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

Preferred report: USENIX OSDI, 2020; report GA-PS004. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal exclusive scheduling
Environment	V10032GB; DellR740; worker32cores/768GB; two GPUs/host
Workload	Local experiment: one GPU; 15 RN50 copies; 16 clients/model; SLO10-500ms
Comparator	Clockwork variants; INFaaS on different hardware
Bounded finding	Fig5 measures goodput satisfying SLO, rather than raw request throughput.
Transfer limit	Cross-hardware INFaaS comparison is indirect; controller/worker distinction.
Source anchor	PDF p10; section6.1; Fig5

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.5 GA-S005: Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

Preferred report: USENIX ATC, 2025; report GA-PS005. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	API remoting + temporal swapping
Environment	Worker48vCPU/384GB; four V10032GB
Workload	Eight inference models; warm-up; CV80ms/BERT200ms SLO
Comparator	Native; remote GPU; PCIe/NVLink swap
Bounded finding	BERT Table4: native42ms, remote43ms, PCIe swap144ms, NVLink swap45ms.
Transfer limit	p98 tail; busy time is not SM occupancy; node scaling separate.

Extraction field	Recorded observation
Source anchor	PDF pp10-11; Table4; sections7.2-7.3

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.6 GA-S006: StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

Preferred report: USENIX ATC, 2024; report GA-PS006. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	API/runtime sandbox virtualization
Environment	Four V10016GB; 28cores/128GB; CUDA10.1; PyTorch1.3; TensorFlow2.12
Workload	Three concurrent inference workflows/GPU
Comparator	MPS; INFless; Astraea; Stream-only
Bounded finding	Table2 evaluates runtime sandbox workflows on shared GPUs.
Transfer limit	Sandbox memory separation is not proof of adversarial security.
Source anchor	PDF p10 onward; section9; Table2

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table

QA	Score	Source-based rationale
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.7 GA-S007: KRYPTON: Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception

Preferred report: USENIX ATC, 2025; report GA-PS007. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	Kernel API virtualization + hybrid
Environment	A10040GB/RTX4090; Ubuntu20.04; kernel5.15.0-91; driver550.90.07; CUDA12.1; PyTorch2.2.1
Workload	Inference/training/rendering in separate containers; solo-throughput targets
Comparator	Temporal; MIG; adapted gpulet
Bounded finding	Inference-specific sharing comparisons; section2.2 tests fault propagation.
Transfer limit	Baseline adaptations and workload mix constrain attribution.
Source anchor	PDF p10 Tables1-2; section5; section2.2 Fig3

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	1	Environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.8 GA-S008: SIRIUS: Colocating ML Inference and Training with Fast GPU Memory Handover

Preferred report: USENIX ATC, 2025; report GA-PS008. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal + MPS memory handover
Environment	V10016GB; Ubuntu20.04; CUDA11.6; PyTorch2.1.2; TVM0.14; LLM case A10080GB
Workload	One GPU; six DNNs/up to56 instances + SWIN training; 300s; Poisson MAF; SLO4x solo
Comparator	Reimplemented TaskSwitch; SP50/SP75; UM+MPS
Bounded finding	Inference p99/SLO outcomes accompany training progress; separate TTFT/TBT LLM case.
Transfer limit	TaskSwitch is a reimplementation, not vanilla PipeSwitch.
Source anchor	Section6.1 Fig10; section6.6

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	1	Measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.9 GA-S009: Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads

Preferred report: ACM ASPLOS Volume 1, 2025; report GA-PS009. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	CUDA API interception + priority
Environment	A10040GB; Ubuntu20.04; CUDA12.2; PyTorch2.2; ONNX1.17
Workload	Same-GPU pairs; MAF inference traces; high-priority p99
Comparator	Time slicing; MPS; priority MPS; TGS
Bounded finding	At 10% idle time: normalized throughput Tally1.19 vs TGS1.65.

Extraction field	Recorded observation
Transfer limit	Latency protection sacrifices some BE throughput; author manuscript.
Source anchor	PDF pp9-11; sections5-5.3; Fig6

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.10 GA-S010: ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments

Preferred report: IEEE/ACM SC, 2024; report GA-P0589. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	MIG + MPS
Environment	A10080GB; Ubuntu20.04; CUDA12.0.1; reported PyTorch1.14.0 needs checking
Workload	Eleven models; six scenario mixes; same-model MPS inside MIG
Comparator	MIG-serving; gpulet and system variants
Bounded finding	Fig7 separates internal/external fragmentation; Fig5 counts GPUs.
Transfer limit	Author manuscript; unusual runtime version; GPU count is not occupancy.
Source anchor	PDF p8 onward; sectionIV; TableIV; Figs5/7

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear

QA	Score	Source-based rationale
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.11 GA-S011: Orion: Interference-aware Fine-grained GPU Sharing for ML Applications

Preferred report: ACM EuroSys, 2024; report GA-PS011. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	CUDA API interception + temporal
Environment	V10016GB/CUDA10.2; A10040GB/CUDA11.3; PyTorch1.12; Python3.9
Workload	Inference-inference and inference-training; uniform/Poisson; three repeats
Comparator	MPS; streams; temporal; REEF-N reimplementation
Bounded finding	Interference-aware scheduling compares inference latency and BE throughput.
Transfer limit	Prototype clients share a process; REEF-N differs from AMD REEF.
Source anchor	PDF pp8-9; sections5.3-6.1; Tables1/3

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	1	Measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.12 GA-S012: GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

Preferred report: ACM SoCC, 2020; report GA-PS012. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	MPS adaptive provisioning
Environment	One V10016GB; XeonGold6148; kernel4.4.0-150; DPDK19.02; CNTK/PyTorch/TensorRT
Workload	Two-four inference functions; VGG case SLO50ms; traffic replay
Comparator	Standalone functions; MPS provisioning variants
Bounded finding	GUE measures efficiency; shadow instance loading takes about12s in Fig15.
Transfer limit	GUE is not GPU busy percentage; median latency not p99; academic mirror.
Source anchor	PDF pp8-12; section4; Table3; Fig15

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	1	Measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.13 GA-S013: KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers

Preferred report: IEEE HPCA, 2023; report GA-P0594. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	Spatial partition/API control (AMD)
Environment	One MI50/60CUs; dualEPYC7302/512GB; Ubuntu18.04; kernel5.4; ROCm4.5
Workload	One/two/four workers and heterogeneous pairs; batch32; aggregate8/16; eight models
Comparator	AMD unrestricted concurrency labelled MPSDefault

Extraction field	Recorded observation
Bounded finding	Kernel-wise right-sizing evaluates throughput and p95 inference latency.
Transfer limit	Feature emulated; overhead subtracted; baseline is not NVIDIA MPS.
Source anchor	PDF pp9-10; sectionsV-VI; TableIII

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	1	Environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	0.5	Partial: interpretability; experiments not rerun

A.14 GA-S014: XSched: Preemptive Scheduling for Diverse XPUs

Preferred report: USENIX OSDI, 2025; report GA-PS014. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal API interposition
Environment	One GV100 inference case; broader XPU experiments separate
Workload	Two BERT-large clients; HP10req/s; saturated BE
Comparator	Triton; Triton+priority; Triton+XSched
Bounded finding	Fig15: relative p99 1.53/1.51/1.07 respectively versus solo.
Transfer limit	Data assumed to fit memory; CUDA graphs and undocumented driver limits.
Source anchor	PDF pp14-15; section8 Case3 Fig15; section9

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear

QA	Score	Source-based rationale
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.15 GA-S015: Paella: Low-latency Model Serving with Software-defined GPU Scheduling

Preferred report: ACM SOSP, 2023; report GA-PS015. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal software scheduling + MPS comparison
Environment	TeslaT4; XeonSilver4114/64GB; Linux4.15.0; CUDA11.7; modified TVM0.10
Workload	Eight models; lognormal sigma1.5/2; stabilized measurements
Comparator	CUDA single/multiple streams; MPS; Triton23.03; ablations
Bounded finding	p99 latency and throughput under bursty model mixes; Fig12 includes MPS.
Transfer limit	Custom Triton TVM wrapper; MPS omitted from Fig11; dynamic batching unsupported.
Source anchor	PDF pp9-12; section7 Table3/Figs11-12; section8

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.16 GA-S016: Optimizing High-Throughput Inference on Graph Neural Networks at Shared Computing Facilities with the NVIDIA Triton Inference Server

Preferred report: Computing and Software for Big Science, 2024; report GA-P0041. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	MIG + concurrent model serving
Environment	A10080GB; 20/40GB slices; two hosts/four GPUs each; AMD Epyc7543; 100Gbps
Workload	Distinct ParticleNet copies within one MIG slice; four requesting workers/model; 120s samples
Comparator	20GB vs40GB slices; scaling settings
Bounded finding	Fig10: thrashing after two models/20GB or five/40GB.
Transfer limit	Local slice evidence only; multi-server scaling separate; isolation properties described, not attack-tested.
Source anchor	HTML Multi-model Scaling/Fig10; Triton implementation; limitations

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.17 GA-S017: VastPipe: A High-Throughput Inference System via Adaptive Space-Division Multiplexing for Diverse Accelerators

Preferred report: Journal of Computer Science and Technology, 2025; report GA-P0006. **Current QA:** 6.5/10 (moderate).

Extraction field	Recorded observation
Mechanism	MPS adaptive quotas; separate AMD CU-mask stratum

Extraction field	Recorded observation
Environment	Local NVIDIA A100: TensorRT serving, MPS; AMD MI100: TVM/ROCm with hipExtStreamCreateWithCUMask. GPU memory, OS, driver and runtime versions NOT EXTRACTED for these GPU experiments. NPU Table3 is not the GPU configuration.
Workload	Co-located CV pairs (VGG16/GoogLeNet, VGG19/GoogLeNet, ResNet101/InceptionV3, VGG19/ResNet50); NLP/CV pairs with BERT/GPT2; Poisson-generated request mixes. Exact GPU batch sizes and rates NOT EXTRACTED.
Comparator	No MPS; default MPS without per-service quotas; RL-selected MPS quotas. AMD no/default CU mask separately.
Bounded finding	A100 VGG16+GoogLeNet: throughput 2.7x no MPS and 1.5x default MPS; not a universal average or p99 result. NLP/CV mean 2.43x no MPS, 1.45x default MPS. MI100 mean 1.89x no mask, 1.24x default mask.
Transfer limit	Include only local GPU inference evidence, not the main 1000-NPU cluster results. Offline Q-table assumes relatively stable request mix; GPU-specific quotas, repetitions and variance unresolved. Chart bars are not preserved in Markdown. Preliminary ICCD2021 report is linked as an unassessed version lead.
Source anchor	Published pp457-459; sections6.7-6.9; Figs14-16; uploaded Markdown lines1150-1290

QA	Score	Source-based rationale
Q1	1	Local model co-location question explicit in6.7/6.8
Q2	1	Per-model MPS control and CU-mask API specified
Q3	0.5	GPU models/backends known; GPU OS/driver/version/memory unresolved
Q4	1	Named co-located models and request distribution; exact load incomplete
Q5	0.5	No/default/tuned sharing compares same platform; exact quotas incomplete
Q6	1	Throughput ratios and requests/s labelled; no tail-latency inference
Q7	0	GPU-specific warm-up, run duration, repetitions and variance not recoverable from supplied text
Q8	0.5	Figures/captions and numerical prose survive; chart bars do not
Q9	0.5	Stable-mix/Q-table assumptions stated; GPU threats incompletely discussed
Q10	0.5	GPU results interpretable with caveats; insufficient configuration for replication

A.18 GA-S018: ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs

Preferred report: CCF Transactions on High Performance Computing, 2023; report GA-P0037.
Current QA: 7.0/10 (moderate).

Extraction field	Recorded observation
Mechanism	CUDA API interception + NVML monitoring and admission throttling
Environment	Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5. Per-GPU memory/framework versions NOT EXTRACTED.
Workload	Five MLPerf inference LC models (ResNet50, SSD-MobileNet, SSD-ResNet34, DLRM, BERT), each paired locally with six BE training models (30 pairs). Single-stream query size1; p99 QoS threshold twice solo. Separate four-GPU app-mixes are not local occupancy evidence.
Comparator	MPS (Non-ArkGPU); k8s-native for QPS/utilization; GaiaGPU and adjustment-unit ablation. MPS quota settings NOT EXTRACTED.
Bounded finding	Section5.2 reports all30 tested pairs satisfy the 2x-solo p99 QoS target with ArkGPU. Low-contention pairs can lose about5% QPS from throttling. Four-GPU app-mix goodput gains94.98% vs k8s-native and38.65% vs MPS are separate system-level results.
Transfer limit	Mixed inference/training qualifies via explicit inference p99/QPS and workload transfer rationale; it does not establish inference-only performance. Polling/API sleeps can add delay; BE checkpoint/restart affects costs. Huge goodput ratios can have near-zero baseline yield. Repetitions/variance and chart coordinates unresolved.
Source anchor	Published pp313-319; sections4.3, 5.1-5.4; Table3; Figs14-21; Markdown lines920-1360

QA	Score	Source-based rationale
Q1	1	LC protection during BE co-location explicit
Q2	1	CUDA interception, NVML polling, sleep and adjustment described in4.3
Q3	1	Table3 specifies GPU/CPU/OS/driver/CUDA/cuDNN; memory not transcribed
Q4	1	MLPerf single-stream, query size1, 30 pairs and QoS rule explicit
Q5	0.5	MPS/native/GaiaGPU compared; quotas and baseline implementation incomplete
Q6	1	p99, average, QPS and yield-based goodput distinguished
Q7	0	Warm-up/window/repetitions/uncertainty not recoverable in supplied text
Q8	0.5	Tables and numerical prose readable; plot data unavailable

QA	Score	Source-based rationale
Q9	0.5	Throttling overhead and low-contention regressions discussed; limited threats analysis
Q10	0.5	Algorithm and data-on-request statement; workload/configuration gaps restrict replication

A.19 GA-S019: Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs

Preferred report: Cluster Computing, 2025; report GA-P0014. **Current QA:** 7.0/10 (moderate).

Extraction field	Recorded observation
Mechanism	MIG dynamic placement/reconfiguration with reconfiguration-last policy
Environment	Two A30s, 24GB each; CUDA12.2; driver535.171.04; DCGM3.1.3. Local sections4.2/4.3.1 use one A30. StaCon profiles2g.12gb+1g.6gb+1g.6gb. OS/backend versions NOT EXTRACTED.
Workload	50 SLM inference jobs; batch1-64; DistilGPT, Codegen, Flan-T5-Large, Codellama and Phi-1 as labelled in Table2. Local batch sensitivity:10 inference jobs, batches1-8 vs32-64 on one A30. Later160-A100 cluster study is simulation.
Comparator	Full-GPU sequential baseline; fixed StaCon MIG; pre-profiled MISO adapted to A30; ORACLE with future job/size/time knowledge.
Bounded finding	Single-A30 batch experiments: smaller partitions increase concurrency but can increase mean JCT; not online p99 evidence. Fig8 broader job set reports39.03% lower makespan vs StaCon, 12.14% vs MISO and14.7% higher normalized STP vs StaCon. Prediction-error sensitivity:15% perturbation adds8.09% makespan.
Transfer limit	Retain one-GPU inference strata separately from two-GPU aggregate and cluster simulation. JCT/makespan/STP cannot replace request-tail latency or goodput. Reconfiguration can stop/restart active jobs; neither architectural MIG claims nor OOM avoidance prove hostile isolation. Model labels/parameter counts and corrupted overhead fraction require caution.
Source anchor	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865

QA	Score	Source-based rationale
Q1	1	Fragmentation/concurrent inference question explicit
Q2	1	MIG profiles, merge/split and placement algorithms described

QA	Score	Source-based rationale
Q3	0.5	GPU/memory/CUDA/driver/DCGM clear; OS/backend unresolved
Q4	1	SLMs, 50 jobs, batch ranges and local10-job strata explicit
Q5	0.5	MISO pre-profiled and adapted; ORACLE privileged information disclosed
Q6	1	JCT/makespan/normalized STP distinct; avoid p99 interpretation
Q7	0.5	Prediction perturbation and power-cap sensitivities; repetitions/variance unresolved
Q8	0.5	Table2 and numeric prose readable; plot coordinates missing
Q9	0.5	Reconfiguration cost/architecture and prediction limits discussed; incomplete threats treatment
Q10	0.5	Implementation described but missing runtime/control detail restricts replication

A.20 GA-S020: iGniter: Interference-Aware GPU Resource Provisioning for Predictable DNN Inference in the Cloud

Preferred report: IEEE Transactions on Parallel and Distributed Systems, 2023; report GA-PX005. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	MPS active-thread quotas + interference-aware batch/provisioning and shadow process
Environment	Ten EC2 p3.2xlarge instances, one V100 each, 8vCPU/61GB; Triton2.12.0, TensorRT8.0.1.6. MPS active-thread percentage; allocation unit2.5%. Heterogeneous extension adds20 T4 instances. OS/driver/CUDA version NOT EXTRACTED.
Workload	AlexNet, ResNet50, VGG19(ImageNet), SSD(VOC2012); 12 workload/SLO configurations; constant arrivals. Local experiment co-locates four models at25% quota, batch3. p99 provisioning measured over windows e.g.30s; model-validation experiments repeated3 times with SD bars.
Comparator	FFD+; GSLICE+ with iGniter placement; gpu-lets+ modified batch strategy and best-fit placement. These are adapted baselines.
Bounded finding	Local four-model latency prediction error1.53%-5.02%(Fig13). Fig14 deployment: iGniter6 GPUs/0 SLO violations vs gpu-lets+8/3, FFD+5/10, GSLICE+6/3; 25% lower hourly instance cost vs gpu-lets+, not25% higher local utilization. Fig17 shows an initial p99 violation before shadow-process recovery.

Extraction field	Recorded observation
Transfer limit	Constant arrivals and four vision models limit bursty/LLM transfer. Negligible PCIe contention assumption can underpredict AlexNet; baselines modified. Steady-state/post-correction SLO success is not an unconditional guarantee; shadow standby incurs capacity/memory costs. Chart bars/latency traces unavailable in Markdown.
Source anchor	Published pp820-824; sections4.2, 5.1-5.4; Figs13-18; Markdown lines938-1245

QA	Score	Source-based rationale
Q1	1	Inference interference and SLO provisioning explicit
Q2	1	Active-thread quotas, batching, placement and shadow recovery described
Q3	0.5	EC2 GPU/CPU/memory and serving versions clear; OS/driver missing
Q4	1	Models, datasets, constant arrivals, quotas and local batch3 clear
Q5	0.5	Three adapted baselines and modifications disclosed; effect confounded with placement/batching
Q6	1	p99, SLO throughput, cost and prediction error distinguished
Q7	1	Model validation three repetitions/SD; p99 windows and update rule explicit; not all experiments have same controls
Q8	0.5	Figure captions/numeric prose survive; error bars and full traces not digitizable
Q9	1	PCIe assumption, model error and recovery described
Q10	1	Public code URL, algorithm and coefficients aid interpretation; code not rerun

A.21 GA-S021: InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems

Preferred report: Future Generation Computer Systems, 2024; report GA-PX006. **Current QA:** 7.5/10 (moderate).

Extraction field	Recorded observation
Mechanism	Temporal SLO-aware scheduling + feedback goodput balancing; MPS comparator
Environment	Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace. OS/driver/CUDA/backend versions NOT EXTRACTED.

Extraction field	Recorded observation
Workload	Local GoogLeNet/ResNet152 Poisson arrivals; three-model test adds Mobile_Pose_MobileNetv3; 600req/s distributed in six ratios; SLO100ms and interval50ms. Separate61-model, 90min MAF replay on four GPUs.
Comparator	Clockwork; MPS plus Clockwork SLO scheduling with two worker processes/GPU; EDF without batching (not a like-for-like batching baseline).
Bounded finding	Local Table3 load3:1:2: unfairness1.05 vs Clockwork1.50; SLO goodput524.29 vs457.24req/s. Other load ratios give smaller differences. Local unfairness is max/min model slowdown(Eq2), not p99 or security. Four-GPU MAF outcome kept separate.
Transfer limit	Two-column Markdown interleaves algorithms/prose; use labelled Table3 rows and sections, not scrambled formulas. Repetitions/variance and software versions unresolved. MAF trace has minute-level invocation counts; cross-study batching and isolation definitions differ.
Source anchor	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899

QA	Score	Source-based rationale
Q1	1	Heterogeneous model-sharing fairness explicit
Q2	1	Estimator, feedback controller and temporal scheduling described
Q3	0.5	Server/GPU/memory clear; OS/driver/backend unresolved
Q4	1	Models, Poisson load, SLO, interval, load ratios and replay duration explicit
Q5	0.5	Clockwork and adapted MPS; EDF lacks batching and is less directly comparable
Q6	1	Goodput requests/s and unfairness ratio distinguish outcome definitions
Q7	0.5	90min replay and parameter sensitivity clear; local repetitions/variance unresolved
Q8	1	Table3 rows/labels unambiguous in Markdown; plot coordinates absent
Q9	0.5	Future work/memory and heterogeneity discussed; incomplete validity treatment
Q10	0.5	Algorithms interpretable despite column interleaving; software/control gaps restrict replication